Electronic Engineering + Software Engineering

DIEGO ARMANDO PINTA CUASQUEN

From sensor to interface.

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PROFILE

I study Electronic Engineering and Software Engineering, and I work right where they meet: complete systems that start at a sensor or a microcontroller and end in an interface where data is seen and controlled.

EXPERIENCE

Engineering intern, Innovation Office, Government of Nariño Jul 2025 – Dec 2025

Designed the enclosure of the NPK device, a soil nutrient sensor, in Autodesk Inventor and took it to a 3D printed prototype.
Repaired the NPK board, replaced its OLED screen and fixed the GPS connection, adding date logging.
Built the NPK page on the office website with a data dashboard, datasheet, manual, request form and role management.
Set up virtual reality headsets with educational apps and staffed the stand at Cumbre CIOS Nariño, La Mejor Taza 2025 and BioExpo 2025.

Engineering intern, Nariño Social Innovation Center (CISNA), ICT Office, Government of Nariño May 2026 – Oct 2026

Designed and built a hybrid solar-electric dryer with automated control for parchment coffee, my degree project.
Pilot trial with 2 paired batches, 221 hours and about 139,800 sensor readings.
Coffee dried 12% faster than in open air, and the harvest point forecast cut the median error from 42.6 h to 7.6 h.
Degree project defended on 5 November 2026 with a Meritorious grade (90/100).

Graphic and motion design, Pasto Noticias Oct 2026

Three editable Photoshop templates for social and broadcast (1920×1080 and 1080×1080), with live text layers, photo clipping masks and contrast fixed to AA.
A vertical 9:16 4K looping animated background for the studio screen, built with 31 openly licensed photos of Pasto and their credits.

Photoshop, Motion graphics, HTML to video, Montserrat

Embedded systems engineer and mobile developer, SISCAN Oct 2026 – present

Smart coffee drying: a hybrid solar-electric system for drying parchment coffee in Nariño, controlled by an ESP32.
Ongoing research: the control system and the mobile app are developed in parallel.

ESP32, Automatic control, Flutter, Dart

Case study Code: github.com/diarpicu2022-commits/siscan-movil

UI designer and frontend developer, NotaScan Oct 2026 – present

Academic grading system: AI reads the handwritten grade, the teacher verifies it and the system stores it.
Complete flow: AI detection, teacher verification and storage.

Tauri, React, TypeScript, Tailwind, Framer Motion

Case study Code: github.com/diarpicu2022-commits/notascan-escritorio

Game developer, Ecos de la Tierra Sep 2026 – present

Turn-based 2D pixel-art RPG about environmental awareness, exportable to PC and Android.
Six playable battles from a region selector, with five characters and the final boss.

Godot 4, GDScript, Python

Case study Code: github.com/diarpicu2022-commits/ecos-de-la-tierra

Android developer, CampusWatch / Agenda Sep 2026 – present

Personal Android planner for people who study and work, with a Wear OS module and encrypted data.
In development, with automated verification on an emulator.

`Kotlin, Jetpack Compose, Wear OS, Room, SQLCipher`

Case study Code: github.com/diarpicu2022-commits/agenda-android

Backend developer, HidroMon Oct 2026

REST API to monitor crop humidity with IoT sensors and threshold alerts.
API documented with an entity-relationship model and examples for every endpoint.

`NestJS, TypeScript, Node.js`

Case study Code: github.com/diarpicu2022-commits/hidromon-nestjs

Content designer and web developer, SETPasto+ Aug 2026

Redesign of Pasto's public transport portal, with a content and voice audit, in four languages.
Redesigned portal with audited content in four languages.

`Node.js, Express, EJS, Tailwind`

Case study Code: github.com/diarpicu2022-commits/Taller-Contenido--voz-y-internacionalizacion

Backend developer, EcoVolt May 2026

Solar energy monitoring and analysis to optimize panel performance.
Foundation for real-time solar production monitoring.

`Python, FastAPI`

Case study Code: github.com/diarpicu2022-commits/EcoVolt

Full-stack IoT developer, AgroPulse Mar 2026 – May 2026

End-to-end greenhouse monitoring: the node reads temperature, humidity and soil, and drives the pump, fan, heater and door.
Complete system across three repositories: firmware, backend and frontend.

`ESP32, C++, PlatformIO, Java 17, Spring Boot, Supabase, Docker, React, TypeScript`

Case study Code: github.com/diarpicu2022-commits/firmware-AgroPulse

Full-stack developer, Harmonia Apr 2026

Smart music player powered by AI.
Full-stack app with authentication, caching and AI integrated.

`Express, TypeScript, MongoDB, Redis, Groq, OAuth, React, Vite`

Case study Code: github.com/diarpicu2022-commits/Harmonia

Backend developer, Proxy IA Apr 2026

AI generation platform using the Proxy pattern to enforce per-plan quotas and rate limits.
The Proxy pattern keeps the business rule (quotas) apart from the model call.

`Java, Spring Boot, React, Docker`

Case study Code: github.com/diarpicu2022-commits/proxy-ai-backend

Game developer, Tower Defense Mar 2026

Tower defense game that uses the Decorator pattern to upgrade towers in real time.
A browser game where every upgrade is a visible decorator.

`Java, Canvas 2D, REST API`

Case study Code: github.com/diarpicu2022-commits/TowerDefenseDecorator

EDUCATION

Software Engineering · fifth semester Ongoing
Universidad Cooperativa de Colombia (UCC)
Software patterns, data structures, software design, interface design and web frameworks.

Electronic Engineering 2021 – 2025
Universidad de Nariño
Specialization track in Automation and Control. Degree project through an internship, a solar-electric coffee dryer with automated control, graded Meritorious (90/100). Graduation ceremony in December 2026.

Technician in Specialty Coffee Production 2020
SENA, Nariño
Technical training in the specialty coffee process. Short SENA courses in computer maintenance (2017), fruit and vegetable post-harvest (2018) and basic English (2022).

SKILLS

Electronics	ESP32, Arduino, GPIO, ADC, DAC, PWM, I^2C, OLED displays, Sensors and actuators
Engineering	MATLAB, Simulink, Automatic control, Modeling and simulation
Languages	C/C++, Java, TypeScript, JavaScript, Python, Kotlin, Dart, GDScript
Backend	Spring Boot, NestJS, Express, FastAPI
Frontend & apps	React, Next.js, Vite, Tailwind CSS, Flutter, Jetpack Compose, Tauri, Godot
Data	PostgreSQL (Supabase), MongoDB, Redis, SQLite, Room
Deployment & tools	Docker, Render, Vercel, Git & GitHub, PlatformIO, Wokwi

LANGUAGES

Spanish	Native
English	English proficiency requirement passed (Universidad de Nariño)